The empirical recurrence for OEIS A207877, proved by transfer matrix

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Abstract

OEIS A207877 counts arrays over $\{0,\ldots,1\}$ of width 6 that avoid the patterns $(0,0,0),(1,0,1)$ along one axis and $(0,1,0),(1,1,1)$ along the other, and carries an empirical recurrence of order 27 contributed by R. H. Hardin. It is true. The patterns have length 3, so the constraint across the growing direction ties together 3 consecutive rows: taking a window of 2 consecutive rows as a state makes the array a walk in a finite digraph on $S=361$ vertices, and $a(n)=\mathbf{1}^\top P^{n-2}\mathbf{1}$. Writing q for the characteristic polynomial of the conjectured recurrence, the residual is $\mathbf{1}^\top P^{n-2-27}q(P)\mathbf{1}$, which vanishes identically beyond a threshold that the computation pins down exactly. Finite, exact, integer.

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1 The sequence and the conjecture

OEIS A207877 is “Number of n X 6 0..1 arrays avoiding 0 0 0 and 1 0 1 horizontally and 0 1 0 and 1 1 1 vertically.”. It has offset 1 and begins

$$19, 361, 702, 1940, 4160, 10742, 24952, \ \ldots$$

The entry carries the following comment:

> Empirical: a(n) = 2*a(n-2) +16*a(n-3) +8*a(n-4) -24*a(n-5) -115*a(n-6) -22*a(n-7) +125*a(n-8) +346*a(n-9) -15*a(n-10) -362*a(n-11) -654*a(n-12) +63*a(n-13) +326*a(n-14) +940*a(n-15) -35*a(n-16) -527*a(n-17) -658*a(n-18) -497*a(n-19) +552*a(n-20) +390*a(n-21) +320*a(n-22) +60*a(n-23) -352*a(n-24) -8*a(n-25) -192*a(n-26) +96*a(n-27) for n>33.

As of the “Last modified” line on the live entry (Jun 05 2026 13:25:11, revision 11) the statement is still recorded as empirical, and nothing on the entry records it as settled either way.

Written out, the claim is

$$a(n) \;=\; 2\,a(n-2)+16\,a(n-3)+8\,a(n-4)-24\,a(n-5)-115\,a(n-6)-22\,a(n-7)+125\,a(n-8)+346\,a(n-9)-15\,a(n \tag{1}$$

for all $n>33$.

2 The array as a walk

Call a row *admissible* if the word it forms contains none of the patterns $(0,0,0),(1,0,1)$ as 3 consecutive entries. There are 19 admissible rows; that condition involves a single row only.

The remaining patterns $(0,1,0),(1,1,1)$ are forbidden along the other axis, so each of them constrains 3 consecutive rows at a fixed position of the column. Take as state a window $w=\big(w^{(1)},\ldots,w^{(2)}\big)$ of 2 consecutive admissible rows, so $S=361$, and declare $w\to w'$ an edge when w' is w with its first entry dropped and one further admissible row t appended, subject to: for every position j, the 3-tuple read down the column j across $w^{(1)}, \dots, w^{(2)}, t$ is none of $(0,1,0), (1,1,1)$. Let P be the adjacency matrix.

Lemma 1. *$a(n) = \mathbf{1}^\top P^{n-2}\mathbf{1}$ for every $n \geq 2$.*

Proof. An array of the entry's kind with n rows is a sequence of n admissible rows in which no forbidden pattern appears along the other axis. Reading it as the sequence of its overlapping windows of length 2 turns it into a walk: consecutive windows overlap in 1 rows by construction, and the edge condition is exactly the requirement on the one new 3-tuple that each step creates. Conversely a walk reassembles into such an array, and the two constructions are mutually inverse. A walk of length $n-2$ visits $n-1$ windows, that is n rows; summing over all starting and ending states gives $\mathbf{1}^\top P^{n-2}\mathbf{1}$. □

3 The criterion

Let $q(t) = t^{27} - \sum_i c_i t^{27-i}$ be the characteristic polynomial of (1) and $u_j = \mathbf{1}^\top P^j q(P)\mathbf{1}$.

Lemma 2. *$a(n) - \sum_i c_i a(n-i) = u_{n-2-27}$ whenever $n - 27 \geq 2$. Hence (1) holds for every $n > t$ exactly when $u_j = 0$ for every $j > t - 2 - 27$. Moreover (u_j) satisfies the monic linear recurrence whose characteristic polynomial is that of P, of degree S, so S consecutive zeros force all later ones and the last nonzero u_j pins t down exactly.*

Proof. Substituting Lemma 1 into the left side and factoring out P^{n-2-27} gives the identity. The equivalence follows by letting the exponent run. For the last claim, Cayley–Hamilton gives $\chi(P) = 0$ for the characteristic polynomial χ of P, so χ applied to the shift annihilates (u_j); being monic of degree S, it propagates S consecutive zeros forward. □

4 The computation

Theorem 3. *The recurrence (1) holds for every $n > 33$. This is the statement of Section 1.*

Proof. The vector $q(P)\mathbf{1} \in \mathbb{Z}^S$ was formed by 27 matrix–vector products in exact integer arithmetic and the u_j evaluated in turn. They vanish from the index corresponding to $n = 33 + 1$ onwards, and the run was continued past S consecutive zeros, which by Lemma 2 settles every larger index. □

5 Verification

Three checks, each independent of the algebra above.

First, the digraph was built directly from the entry's stated patterns and the walk counts compared against the entry's DATA at every one of the 25 published indices; the values agree exactly, including the initial count of 19 admissible rows. That is what ties the digraph to the sequence, and a model reproducing only part of the published data was rejected wherever it occurred.

Second, the recurrence (1) was evaluated on those published terms in exact integer arithmetic, with no matrices involved, and vanishes at every index the entry claims.

Third, the vanishing test of Lemma 2 was carried past $S = 361$ consecutive zeros rather than to a sampled prefix.

References

[1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, https://oeis.org, 2026. Sequence A207877.

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[4] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.